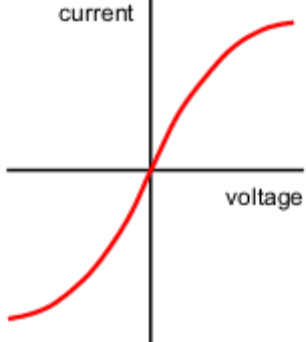


6(a)(i)	$R = \rho L / A$ $P = I^2 R$ Therefore rate of heat loss, $P = \rho L / A$ $P_X / P_Y = (\rho_X / \rho_Y) \times (L_X / L_Y) \times (A_Y / A_X)$ $= (1/1.58) (1/1.50) (1.50)$ $= 0.633$	 [1] [1]
(ii)	Wire X because there is lower rate of energy loss as thermal energy in the wire.	[1]
(iii)1.	The cable is made up of 5 thin wires in parallel. Let resistance of a single thin wire be R. $1 / 0.0458 = 5 (1 / R)$ $R = 0.229 \Omega$	 [1] [1]
2.	Any one: Several thin wires are more flexible than a single thick wire Thin wires have more surface area and thus dissipate heat more effectively/better dissipate the heat (Note: both wires have the same heat production but heat dissipation from the wire can be different) Cable remains workable even when one wire is broken	[1]
(b)(i)	 Note: The line must not be horizontal at high voltage. Hence do not overdo the decreasing slope. The graph near the origin should be linear.	[1]
(ii)	With increasing magnitude of V, more heat is dissipated and temperature of filament rises, Lattice metal ions vibration amplitude increase, Collision frequency between metal ions and electrons increase, Resistance increases (and the current increases at a decreasing rate.)	 [1] [1] [1]

(iii)	No, the equation is only true where the I-V graph is a straight line passing through origin (ohmic conductor) but this I-V graph is not. OR No, as the resistance is represented by the inverse of the gradient of a line connecting that point to the origin. It is not the inverse of the tangent at that point because the resistance is the ratio of the potential difference to the current.	[1]
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7(a)(i)	Nuclear fusion is the process where two light/small nuclei are combined	[1]
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